Unit 2 - Linear Algebra
2.5 Determinants and Inverses

Graham West

MSDS 520: Math and Stats Foundations for Data Science
Meharry Medical College

Outline

1. Why Determinants and Inverses?
2. The Determinant as Area
3. When the Determinant Is Zero
4. The Inverse
5. Solving Systems with the Inverse
6. A First Look at Eigenvectors
7. Review

Why Determinants and Inverses?

A Problem to Motivate Us

Back to the pharmacy (Unit 2-1)

A 10 mg/mL stock and a 2 mg/mL stock. Tonight's order: 500 mL at 4 mg/mL. As one matrix equation:

$$\begin{bmatrix} 1 & 1 \\ 10 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 500 \\ 2000 \end{bmatrix}$$

- Row 1: total volume, mL.
- Row 2: total drug, mg.

But the orders keep coming

order	**mL**	**mg/mL**	**mg**
1	500	4	2000
2	300	6	1800
3	1000	3	3000

Same left side every time. Only the right-hand side changes.

Two questions this lecture answers

1. Will **every** order have exactly one answer? $\Rightarrow$ the **determinant**
2. Can we do the work **once** and reuse it for every order? $\Rightarrow$ the **inverse**

Where This Shows Up in Data Science

Determinants and inverses underneath the tools

- **Fitting a regression.** The best-fit coefficients are $(X^\top X)^{-1} X^\top \bar{y}$: an inverse. Unit 3 builds this.
- **Redundant features.** When two columns of a data table carry the same information (**collinearity**), a determinant goes to 0 and the coefficients become unstable.
- **Unusual lab results.** The Mahalanobis distance, which flags a patient whose values are odd **in combination**, uses the inverse of a covariance matrix.
- **The bell curve in many dimensions.** The multivariate normal's formula contains both $\det \Sigma$ and Σ^{-1}.
- **Eigenvectors.** Google's PageRank and PCA, a standard way to compress data, are both built on them. Today's last section introduces them.

What We Will Build

The plan

1. The **determinant** as an **area scale factor**: the formula $ad - bc$ and where it comes from; **volume** in 3D.
2. Its **sign** (orientation), and its link to the **dot product**.
3. When $\det = 0$: collapse, dependent columns, and Unit 2-1's three outcomes.
4. The **inverse**: undoing a transformation, the 2×2 formula, and the rules.
5. **Solving** $A\bar{x} = \bar{b}$ as $\bar{x} = A^{-1}\bar{b}$, and Python.
6. A first look at **eigenvectors**: the directions that do not turn, found with $\det(A - \lambda I) = 0$.

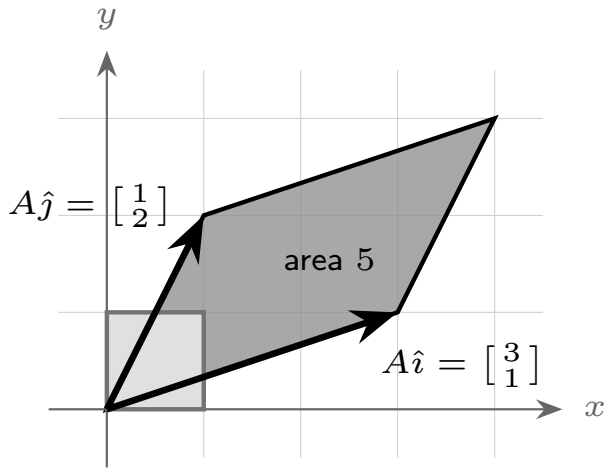
Notebooks for this lecture: `U2-5_Determinants-1` and `-2`; for the eigen section, `U2-6_Eigen-1` and `-2`.

The Determinant as Area

How Much Does a Matrix Stretch Area?

Recall from Unit 2-4

- The columns of A are where $\hat{\imath}$ and $\hat{\jmath}$ land.
- So the **unit square** (area 1) lands on the **parallelogram** the two columns span.
- **Question:** what is that parallelogram's area?



Definition: the determinant

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

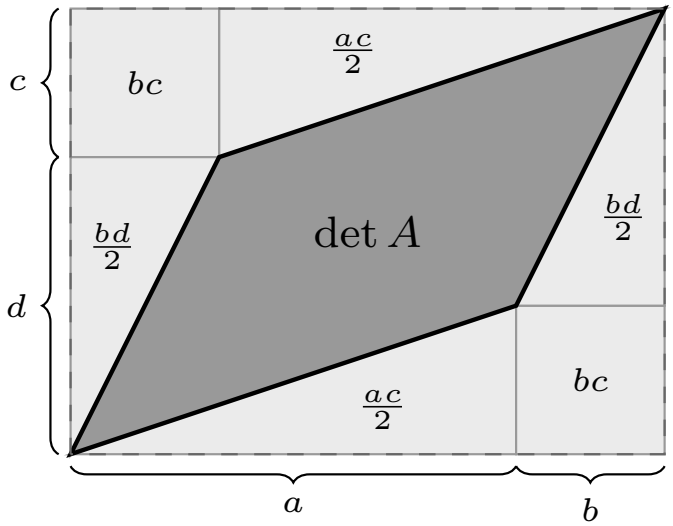
It is the (signed) factor by which A scales **any** area. The unit square is just the easiest example.

Worked: $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

$\det A = 3 \cdot 2 - 1 \cdot 1 = 5$.
Every area is multiplied by 5.

Notebook `U2-5_Determinants-1_AsArea`, §1

Why $ad - bc$? The Box Minus the Corners



Drawn for $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right]$.

Area of the parallelogram

- The enclosing box is $(a + b)$ wide and $(c + d)$ tall.
- Cut off two triangles of $\frac{ac}{2}$, two of $\frac{bd}{2}$, and two rectangles of bc.

$$\begin{aligned} \text{area} &= (a+b)(c+d) - ac - bd - 2bc \\ &= ac + ad + bc + bd - ac - bd - 2bc \\ &= ad - bc \end{aligned}$$

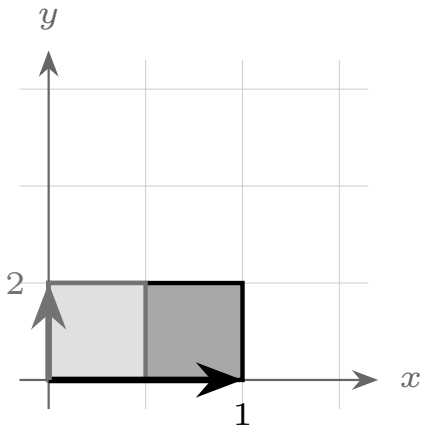
Check with $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right]$

Box $4 \times 3 = 12$. Corners $3 + 2 + 2 = 7$.
Area $12 - 7 = 5 = \det A$.

The picture has every entry positive, but the formula $ad - bc$ holds for every 2×2 matrix.

Parallelograms and Their Determinants

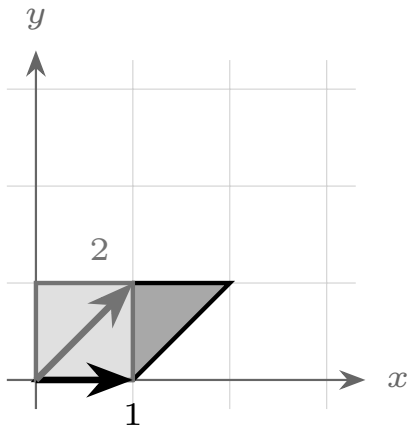
The unit square and its image under four matrices, all drawn to the same scale. Each arrow is labelled with its column number.



$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$\det = 2 - 0 = 2$

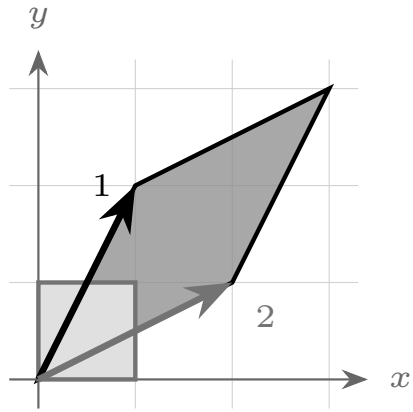
Area **doubles**.



$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$\det = 1 - 0 = 1$

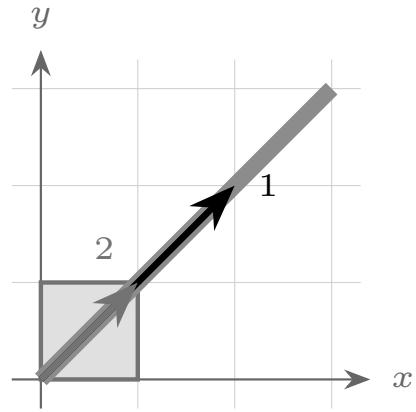
Tilted: area **kept**.



$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$\det = 1 - 4 = -3$

Area $\times 3$, **flipped**.



$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$\det = 2 - 2 = 0$

Flat: area 0.

Size of $\det$**:** the area. **Sign:** the orientation (next slides).

Determinants of the Transformations You Know

Transformation	**Matrix**	$\det$	**What happens to area**
Scaling	$\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$	$2 \cdot \frac{1}{2} = 1$	kept
Uniform scaling	$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$	k^2	multiplied by k^2
Shear	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	1	kept: same base and height
Rotation	$R(\theta)$	$\cos^2\theta + \sin^2\theta = 1$	kept
Reflection	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	-1	kept, but **flipped**
Projection	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	0	crushed to 0

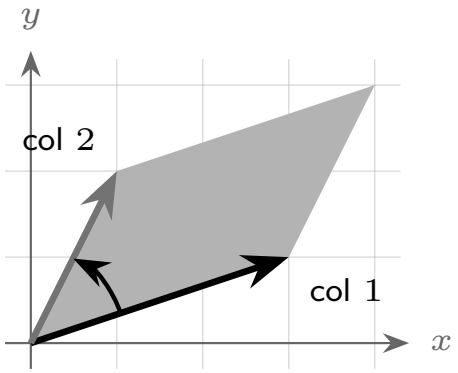
Two moves in a row: the factors multiply

$$\det(AB) = \det A \cdot \det B$$

- B multiplies area by $\det B$, then A multiplies **that** by $\det A$.
- **Example:** rotate, then double everything. $\det = 1 \cdot 4 = 4$. Check: $\det \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = 0 + 4 = 4$.

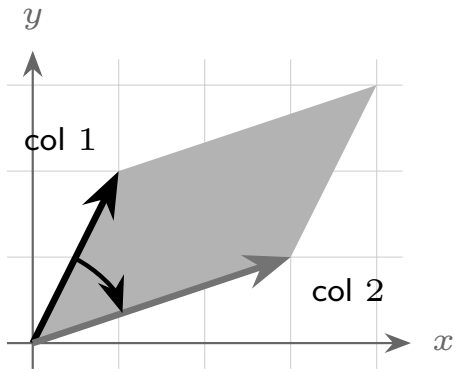
The Sign: Orientation

Same two columns, opposite order. Same parallelogram, same area 5: only the **sign** changes.



$$\det \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} = 6 - 1 = +\mathbf{5}$$

Counterclockwise



$$\det \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} = 1 - 6 = -\mathbf{5}$$

Clockwise

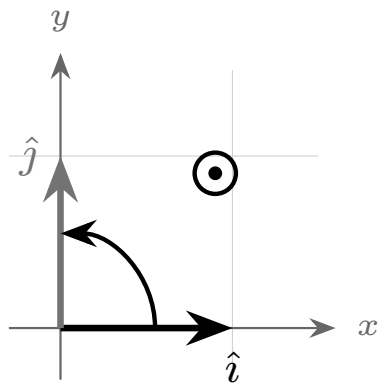
In general: swapping the columns flips the sign

$$\det \begin{bmatrix} b & a \\ d & c \end{bmatrix} = bc - ad = -(ad - bc) = -\det \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The Sign: the Right-Hand Rule

The rule

1. Point your right hand's fingers along **column** 1.
2. Curl them toward **column** 2, the short way round.
3. Thumb **out of** the screen ($\odot$): $\det > 0$. Thumb **into** it ($\otimes$): $\det < 0$, **mirrored**.



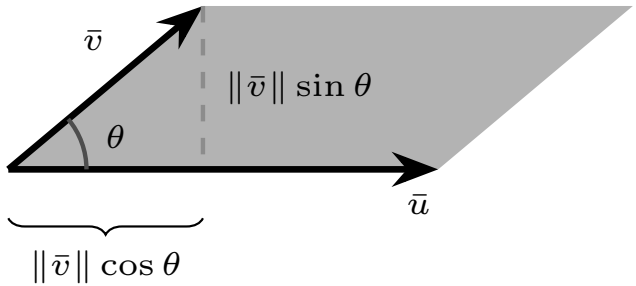
$\hat{\imath}$ to $\hat{\jmath}$: thumb **out**, along the z-axis.

Check it on matrices you know

- $I = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$: columns $\hat{\imath}$ then $\hat{\jmath}$, thumb out, $\det I = +1$.
- Reflection $\left[\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$: from $\left[\begin{smallmatrix} -1 \\ 0 \end{smallmatrix}\right]$ to $\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$ is clockwise, thumb in, $\det = -1$.
- In 3D the thumb **is** the third axis: $\det > 0$ when column 3 sits on the thumb's side.

Notebook `U2-5_Determinants-1_AsArea`, §3, the sign

The Determinant and the Dot Product



Shadow along $\bar{u}$: the cosine. Height off $\bar{u}$: the sine.

Dot product: the **parallel** part

$$\bar{u} \cdot \bar{v} = \|\bar{u}\| \, \|\bar{v}\| \cos\theta$$

How much of $\bar{v}$ lies **along** $\bar{u}$. Zero when they are **perpendicular**.

Determinant: the **perpendicular** part

$$\det \begin{bmatrix} \bar{u} & \bar{v} \end{bmatrix} = \|\bar{u}\| \, \|\bar{v}\| \sin\theta$$

Base $\|\bar{u}\|$ times height $\|\bar{v}\| \sin\theta$. Zero when they are **parallel**.

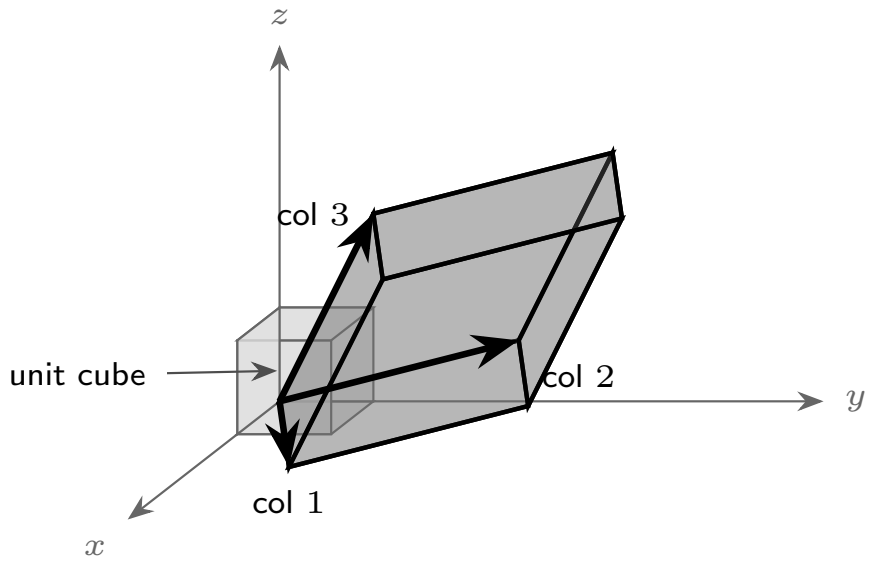
So: $\det = 0$ exactly when the columns are parallel

Parallel means one column is a multiple of the other: the columns are **linearly dependent** (Unit 2-2). (Here θ is measured counterclockwise from $\bar{u}$ to $\bar{v}$, which is what lets the sine carry the sign.)

Bigger Matrices: Volume and Beyond

Same meaning, more dimensions

- 2×2: the **area** factor.
- 3×3: the **volume** factor. The unit cube becomes a slanted box, a **parallelepiped**.
- $n \times n$: the **hyper-volume** factor.
- Sign: orientation. $\det = 0$: flattened.



The box spanned by the columns of $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$.

But the formula gets cumbersome

- Terms in the formula: 2 for 2×2, 6 for 3×3, 24 for 4×4, over 3.6 million for 10×10.
- So computers find $\det$ by **elimination** instead.

Exercise: Determinants

Try it

1. Compute each determinant and say what the matrix does to area:

$$\begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}, \qquad \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}.$$

2. A triangle of area 6 is transformed by $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$. What is the area of its image?

3. For which values of k does $\begin{bmatrix} k & 4 \\ 1 & k \end{bmatrix}$ squash the plane flat?

Exercise: Solution

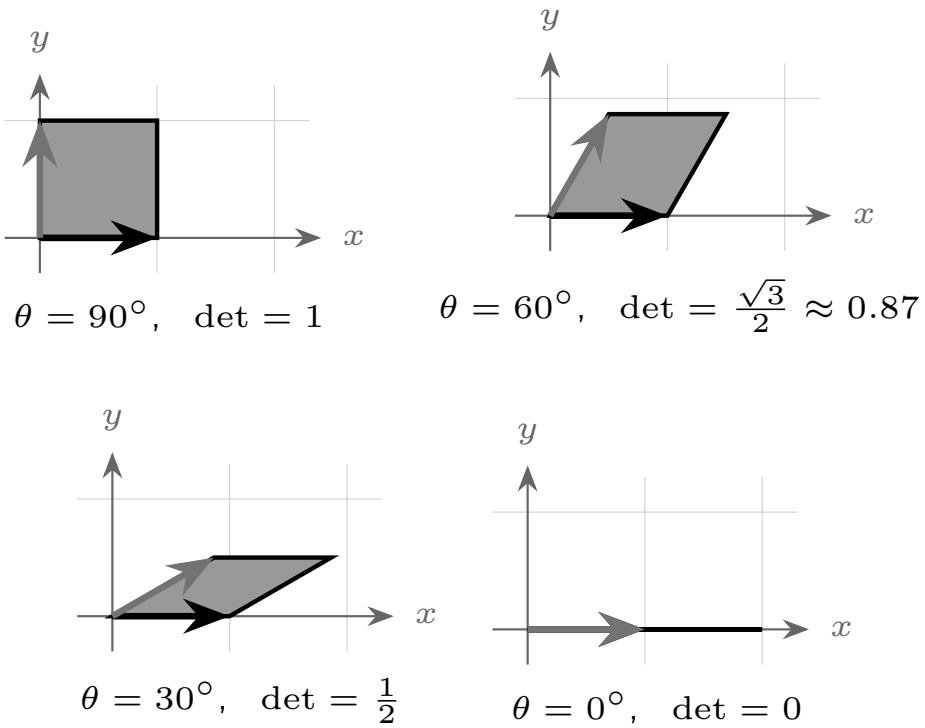
Answers

1. The four area factors:
   - $4 \cdot 3 - 0 \cdot 0 = 12$: x is stretched by 4 and y by 3, so area is multiplied by 12.
   - $2 \cdot 3 - 5 \cdot 1 = 1$: the shape is distorted, but its area is **kept**.
   - $0 \cdot 0 - 2 \cdot 3 = -6$: area times 6, and **flipped**. Here $\hat{\imath} \mapsto \left[\begin{smallmatrix} 0 \\ 3 \end{smallmatrix}\right]$ and $\hat{\jmath} \mapsto \left[\begin{smallmatrix} 2 \\ 0 \end{smallmatrix}\right]$: the columns now turn clockwise.
   - $2(-2) - (-1)(4) = -4 + 4 = 0$: area **crushed to** 0. Column 2 is $-\frac{1}{2}$ times column 1.
2. Every area is multiplied by $\det = 3 \cdot 2 - 1 \cdot 1 = 5$, so the image has area $6 \times 5 = 30$. The triangle's shape does not matter.
3. Flat means area 0: $\det = k \cdot k - 4 \cdot 1 = k^2 - 4 = 0$, so $k = 2$ or $k = -2$. Check $k = 2$: $\left[\begin{smallmatrix} 2 & 4 \\ 1 & 2 \end{smallmatrix}\right]$ has column $2 = 2\times$ column 1.

When the Determinant Is Zero

Watching the Area Drain Away

Hold column 1 at $\left[\begin{smallmatrix}1\\0\end{smallmatrix}\right]$ and swing column 2, also a unit vector, down toward it: $A(\theta) = \left[\begin{smallmatrix}1 & \cos\theta\\ 0 & \sin\theta\end{smallmatrix}\right]$, so $\det A(\theta) = \sin\theta$.



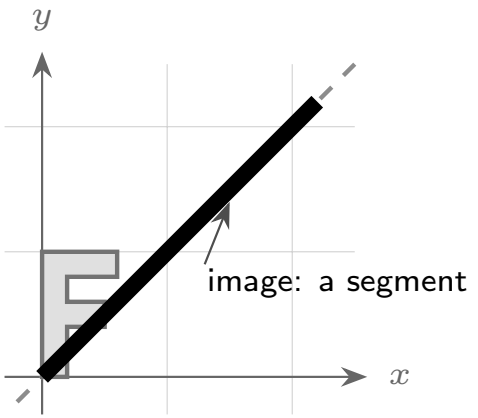
- As the columns line up, the parallelogram **flattens** and its area drains to 0.
- At $\theta = 0°$ the whole plane lands on the x-axis: a **collapse** to a line.

Notebook `U2-5_Determinants-1_AsArea`, §2

$\det = 0$: Three Ways of Saying One Thing

$$M = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\det M = 2 \cdot 1 - 1 \cdot 2 = 0$$



M lands the whole F on the line $y = x$.

One fact, three statements

1. **Geometric: zero area.** M squashes the whole plane onto one line, $y = x$.
2. **Redundancy: the columns are linearly dependent** (Unit 2-2). Column $1 = \left[\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}\right]$ is $2\times$ column $2 = \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$.
3. **Systems: never exactly one solution** (Unit 2-1). Every $M\bar{x}$ lies on $y = x$, so $M\bar{x} = \bar{b}$ has
   - **no** solutions if $\bar{b}$ is off the line ($\left[\begin{smallmatrix} 3 \\ 5 \end{smallmatrix}\right]$);
   - **infinitely many** if $\bar{b}$ is on it ($\left[\begin{smallmatrix} 3 \\ 3 \end{smallmatrix}\right]$).

Definition: singular

A square matrix with $\det = 0$ is **singular**. With $\det \neq 0$ it is **nonsingular**.

Notebook `U2-5_Determinants-1_AsArea`, §4, $\det = 0$ checked three ways

Read the Outcome Before Solving

For $ax + by = e, \quad cx + dy = f$: look at $\det \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$

- $\det \neq 0$: the lines cross. **Exactly one solution**, for **every** right-hand side.
- $\det = 0$: same slope, since $\frac{a}{b} = \frac{c}{d}$ means $ad = bc$. **None or infinitely many.**
- $\det$ sees only the coefficients, so it cannot tell "none" from "infinitely many". Elimination can.

Question 1, answered: the pharmacy

$$x + y = 500, \quad 10x + 2y = 2000 \qquad \Rightarrow \qquad \det \begin{bmatrix} 1 & 1 \\ 10 & 2 \end{bmatrix} = 2 - 10 = -8 \neq 0$$

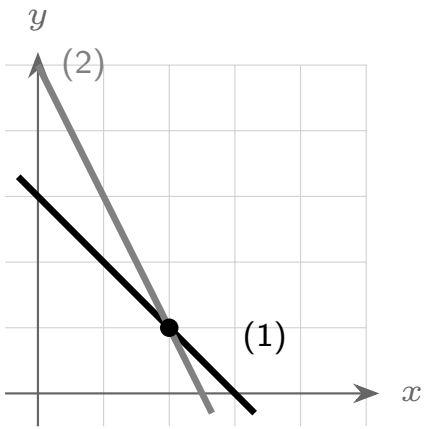
The left side never changes from order to order, so **every** order has exactly one answer.

Next slide: three systems, plotted, with their determinants.

Three Systems, Plotted

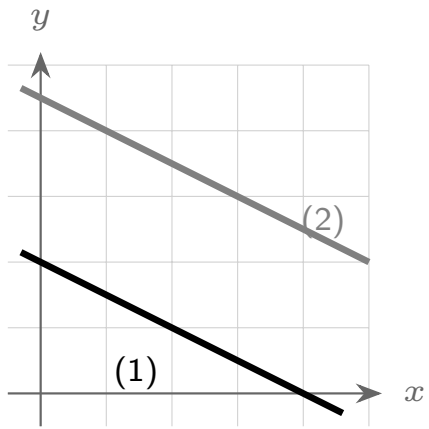
Each line is labelled with its equation's number.

$$x + y = 3$$
$$2x + y = 5$$
$$\det \left[\begin{smallmatrix} 1 & 1 \\ 2 & 1 \end{smallmatrix}\right] = 1 - 2 = -1$$



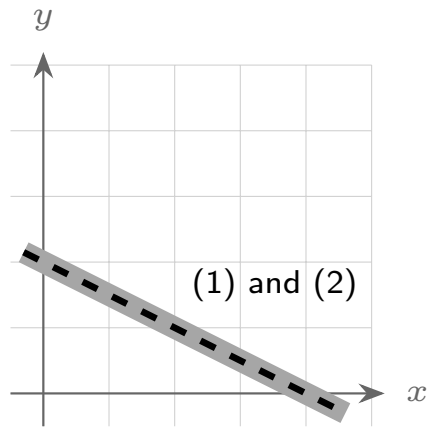
$\det \neq 0$: they **cross**.
Exactly one: $(2, 1)$.

$$2x + 4y = 8$$
$$x + 2y = 9$$
$$\det \left[\begin{smallmatrix} 2 & 4 \\ 1 & 2 \end{smallmatrix}\right] = 4 - 4 = 0$$



$\det = 0$: **parallel**.
No solution.

$$2x + 4y = 8$$
$$x + 2y = 4$$
$$\det \left[\begin{smallmatrix} 2 & 4 \\ 1 & 2 \end{smallmatrix}\right] = 4 - 4 = 0$$



$\det = 0$: the **same line**.
Infinitely many.

Systems 2 and 3 share a coefficient matrix, so they share $\det = 0$. Only the right-hand side tells them apart.

Exercise: Zero or Not?

Try it

1. Without solving, decide whether each system has exactly one solution. If it does, find it; if not, say whether it has none or infinitely many.
   (a) $3x - y = 1, \quad 6x - 2y = 4$ (b) $x + 2y = 5, \quad 3x - y = 1$
   (c) $0.5x + y = 2, \quad x + 2y = 4$
2. The columns of a 3×3 matrix are $\bar{u}$, $\bar{v}$, and $\bar{u} + \bar{v}$. What is its determinant, and why?

Exercise: Solution

Answers

1. (a) $\det = 3(-2) - (-1)(6) = -6 + 6 = 0$: not exactly one. The left side of equation 2 is twice equation 1's, but $4 \neq 2 \times 1$: parallel lines, **no solution**.

   (b) $\det = 1(-1) - 2(3) = -7 \neq 0$: **exactly one**. Elimination gives $(x, y) = (1, 2)$. Check: $1 + 4 = 5$ and $3 - 2 = 1$.

   (c) $\det = 0.5 \cdot 2 - 1 \cdot 1 = 0$. Equation 2 is exactly twice equation 1, right side included: the same line, **infinitely many**.

2. $\det = 0$. The third column is a combination of the first two, so it lies in their plane. All three columns span only a plane, the slanted box is flat, and its volume is 0.

The Inverse

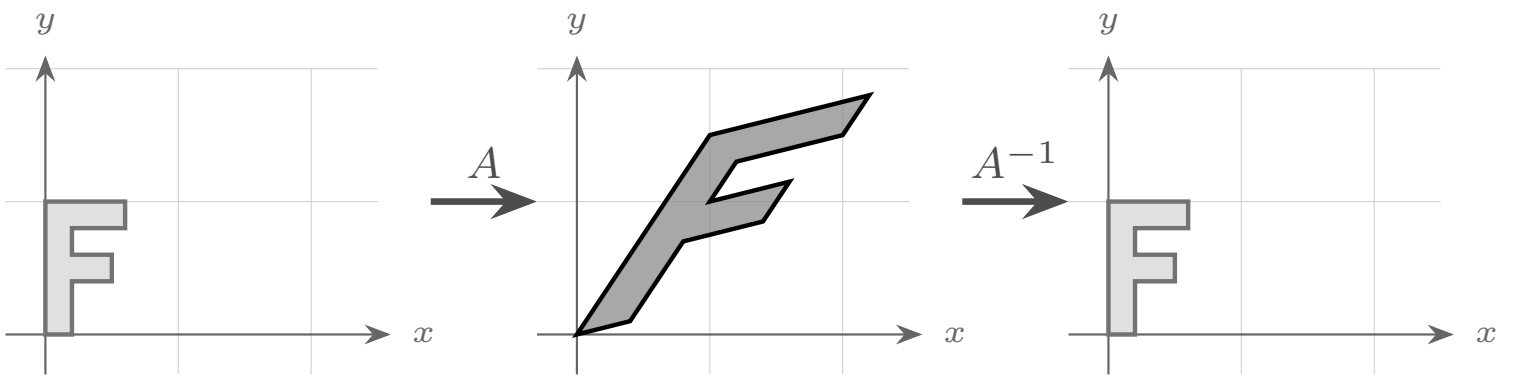
Undoing a Transformation

Definition: the inverse

The **inverse** of a square matrix A is the matrix A^{-1} with

$$A^{-1}A = I \qquad \text{and} \qquad AA^{-1} = I.$$

Apply A, then A^{-1}, and every point is back where it started. A matrix with an inverse is **invertible**; one without is **singular**.



Here $A = \begin{bmatrix} 2 & 1 \\ 0.5 & 1.5 \end{bmatrix}$. Only a **square** matrix can have an inverse.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §1, there and back again

Inverses You Already Know (Unit 2-4)

Move	**Matrix**	**How to undo it**	**Inverse**
Scaling	$\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$	halve x, double y	$\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{bmatrix}$
Shear	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	slide back the other way	$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$
Rotation	$R(\theta)$	turn back	$R(-\theta)$
Reflection	F	flip again	F itself
Projection	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	**impossible**	none

Check one

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Why projection cannot be undone

$(3, 5)$, $(3, -2)$, and $(3, 0)$ all land on $(3, 0)$. Nothing can know which one to send it back to.

Numbers do the same: $5 \cdot \frac{1}{5} = 1$, and 0 has no reciprocal. For matrices, the role of 0 is played by **every** matrix with $\det = 0$.

The 2×2 Inverse Formula

The formula

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Swap the diagonal, **negate** the off-diagonal, **divide** by $\det A$.

Step 1: swap and negate. This **almost** undoes A

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{bmatrix} = \begin{bmatrix} \det A & 0 \\ 0 & \det A \end{bmatrix} = (\det A)\, I$$

Back to I, except everything is still scaled by the factor $\det A$.

Step 2: divide by $\det A$. This removes the leftover scaling

- A multiplies every area by $\det A$, so the undo must multiply every area by $1/\det A$. Indeed $\det(A^{-1}) = 1/\det A$.
- If $\det A = 0$ there is nothing to divide by: a shape squashed to area 0 cannot be scaled back up.

Worked: a 2×2 Inverse

$A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right]$

1. **Determinant:** $\det A = 3 \cdot 2 - 1 \cdot 1 = 5 \neq 0$, so the inverse exists.
2. **Swap, negate, divide:**

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$$

3. **Check:**

$$AA^{-1} = \frac{1}{5} \begin{bmatrix} 3(2)+1(-1) & 3(-1)+1(3) \\ 1(2)+2(-1) & 1(-1)+2(3) \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = I$$

The area bookkeeping

A multiplies area by 5, and $\det(A^{-1}) = \frac{1}{25}(2 \cdot 3 - (-1)(-1)) = \frac{5}{25} = \frac{1}{5}$: the undo divides it back by 5.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §2, the formula built by hand

Why the Determinant Sits in the Denominator

At zero: $\det A = 0$ makes the formula divide by zero. A **is invertible exactly when** $\det A \neq 0$**.**

Near zero: a matrix whose determinant is a small ε

$$M = \begin{bmatrix} 1 & 1 \\ 1 & 1+\varepsilon \end{bmatrix}$$

$$\det M = (1+\varepsilon) - 1 = \varepsilon$$

$\varepsilon = \det M$	M	M^{-1}
1	$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$
0.1	$\begin{bmatrix} 1 & 1 \\ 1 & 1.1 \end{bmatrix}$	$\begin{bmatrix} 11 & -10 \\ -10 & 10 \end{bmatrix}$
0.01	$\begin{bmatrix} 1 & 1 \\ 1 & 1.01 \end{bmatrix}$	$\begin{bmatrix} 101 & -100 \\ -100 & 100 \end{bmatrix}$

- M barely changes; M^{-1} **explodes**.
- **Why data science cares:** nearly dependent (collinear) features do this, so a tiny change in the data makes regression coefficients jump.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §2

Rules Worth Knowing

Determinants

- $\det(AB) = \det A \cdot \det B$
- $\det(A^{-1}) = 1/\det A$
- $\det(A^\top) = \det A$
- $\det(kA) = k^n \det A \quad (n \times n)$

Inverses

- $(A^{-1})^{-1} = A$
- $(AB)^{-1} = B^{-1}A^{-1}$: **reverse order**
- $(A^\top)^{-1} = (A^{-1})^\top$
- $(kA)^{-1} = \frac{1}{k}A^{-1} \quad (k \neq 0)$

Area factors multiply; the undo divides the area back; kA scales **all** n directions by k, so k^2 for 2×2 and k^3 for 3×3.

Socks and shoes: why the order reverses

Socks on, then shoes on. To undo it, shoes off **first**. Check:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Watch out: $\det(A + B) \neq \det A + \det B$. Example: $\det(I + I) = \det(2I) = 4$, but $\det I + \det I = 2$.

Exercise: Inverses

Try it

1. Find the inverse of $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$ and check it by multiplying.
2. Use the formula to invert the quarter turn $R(90°) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Which rotation is the answer?
3. Which of these have an inverse? For any that does, find it.

$$\begin{bmatrix} 4 & 6 \\ 2 & 3 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 3 & 4 \\ 1 & 3 \end{bmatrix}.$$

4. A and B are 2×2 with $\det A = 3$ and $\det B = -2$. Find $\det(AB)$, $\det(A^{-1})$, and $\det(2A)$.

Exercise: Solution

Answers

1. $\det = 2 \cdot 3 - 5 \cdot 1 = 1$, so swap, negate, divide by 1: $\left[\begin{smallmatrix} 3 & -5 \\ -1 & 2 \end{smallmatrix}\right]$. Check: $\left[\begin{smallmatrix} 2 & 5 \\ 1 & 3 \end{smallmatrix}\right]\left[\begin{smallmatrix} 3 & -5 \\ -1 & 2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 6-5 & -10+10 \\ 3-3 & -5+6 \end{smallmatrix}\right] = I$. A determinant of 1 is why there are no fractions.

2. $\det = 0 \cdot 0 - (-1)(1) = 1$. Swap the diagonal (two zeros), negate the off-diagonal: $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$. That is $R(-90°)$, since $\cos(-90°) = 0$ and $\sin(-90°) = -1$: a quarter turn **clockwise** undoes a quarter turn counterclockwise.

3. Check each determinant first:
   - $4 \cdot 3 - 6 \cdot 2 = 0$: **singular**. Column 2 is $1.5\times$ column 1.
   - $0 \cdot 0 - 1 \cdot 1 = -1$: invertible. The formula gives $\frac{1}{-1}\left[\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$: the reflection across $y = x$ is **its own inverse**.
   - $3 \cdot 3 - 4 \cdot 1 = 5$: invertible, with inverse $\frac{1}{5}\left[\begin{smallmatrix} 3 & -4 \\ -1 & 3 \end{smallmatrix}\right]$.

4. $\det(AB) = 3 \cdot (-2) = -6$. $\det(A^{-1}) = \frac{1}{3}$. $\det(2A) = 2^2 \cdot 3 = 12$.

Solving Systems with the Inverse

One System, Four Ways to Write It

The same system, in the order we met each form

Form	**The system**	**What it shows**
System of equations Unit 2-1	$\begin{cases} x_1 + x_2 = 3 \\ x_2 = 2 \end{cases}$	Each equation is one fact, and one line.
Augmented matrix Unit 2-1	$\left[\begin{array}{cc\|c} 1 & 1 & 3 \\ 0 & 1 & 2 \end{array}\right]$	Just the numbers, ready for row operations.
Combination of vectors Unit 2-2	$x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$	The unknowns are the weights on the columns.
Matrix equation Unit 2-3	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$	The whole system as one equation.

The form this section works with: $A\bar{x} = \bar{b}$

- A is the **coefficient matrix**, $\bar{x}$ the **unknowns**, $\bar{b}$ the **right-hand side**.
- The inverse acts on the **matrix equation**: it undoes A to leave $\bar{x}$.

The Methods We Have So Far

Ways to solve a system (Unit 2-1)

Method	**How it works**	**Best when**
Graphing	Draw each line; the solution is where they **cross**.	Picturing it; 2 variables only.
Substitution	Solve one equation for a variable; **plug it into** the rest.	A variable is already isolated.
Elimination	Scale and add equations so a variable **cancels**.	The coefficients are awkward.
Gaussian elimination	Row operations on $[\,A \mid \bar{b}\,]$ to row-echelon form, then back-substitute.	Any size; fully mechanical.
`np.linalg.solve`	The computer runs elimination for you.	Too big to do by hand.

What none of them does

- Every one starts over from scratch when $\bar{b}$ changes, even if A stays the same.
- **This section:** do the work on A **once**, with A^{-1}, and reuse it for every $\bar{b}$.

Solving $A\bar{x} = \bar{b}$ in One Move

Multiply both sides on the left by A^{-1}

$$\begin{aligned} A\bar{x} &= \bar{b} \\ A^{-1}A\bar{x} &= A^{-1}\bar{b} \\ I\bar{x} &= A^{-1}\bar{b} \\ \bar{x} &= A^{-1}\bar{b} \end{aligned}$$

Watch out

- On the **left**, both sides. $\bar{b}A^{-1}$ is not even defined: 2×1 times 2×2.
- For numbers, $3x = 12$ gives $x = \frac{1}{3} \cdot 12$. Matrices have no division; multiplying by A^{-1} takes its place.

Worked: Unit 2-1's system $x + y = 3, \quad 2x + y = 5$

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \qquad A^{-1} = \frac{1}{-1}\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$$

$$\bar{x} = A^{-1}\bar{b} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -3+5 \\ 6-5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

The same $(2, 1)$ that Gaussian elimination found.

Back to the Pharmacy

Step 1: invert the coefficient matrix, once

$$A = \begin{bmatrix} 1 & 1 \\ 10 & 2 \end{bmatrix}, \quad \det A = 2 - 10 = -8, \quad A^{-1} = \frac{1}{-8}\begin{bmatrix} 2 & -1 \\ -10 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{8} \\ \frac{5}{4} & -\frac{1}{8} \end{bmatrix}$$

Step 2: order 1, 500 mL at 4 mg/mL

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{8} \\ \frac{5}{4} & -\frac{1}{8} \end{bmatrix} \begin{bmatrix} 500 \\ 2000 \end{bmatrix} = \begin{bmatrix} -125 + 250 \\ 625 - 250 \end{bmatrix} = \begin{bmatrix} 125 \\ 375 \end{bmatrix}$$

- Draw 125 mL of the strong stock and 375 mL of the weak one.
- Check: $125 + 375 = 500$ mL, and $10(125) + 2(375) = 1250 + 750 = 2000$ mg.

The same answer as Unit 2-1. The difference shows up on the next order.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §3, solving a system with the inverse

Solve Once, Reuse for Every Order

Step 1 is already done: the inverse from the last slide

$$A^{-1} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{8} \\ \frac{5}{4} & -\frac{1}{8} \end{bmatrix}$$

Step 2 again, for orders 2 and 3: one multiplication each

Order	$\bar{b}$	$A^{-1}\bar{b}$	**Check (mg)**
300 mL at 6	$\begin{bmatrix} 300 \\ 1800 \end{bmatrix}$	$\begin{bmatrix} -75+225 \\ 375-225 \end{bmatrix} = \begin{bmatrix} 150 \\ 150 \end{bmatrix}$	$1500 + 300 = 1800$
1000 mL at 3	$\begin{bmatrix} 1000 \\ 3000 \end{bmatrix}$	$\begin{bmatrix} -250+375 \\ 1250-375 \end{bmatrix} = \begin{bmatrix} 125 \\ 875 \end{bmatrix}$	$1250 + 1750 = 3000$

Only $\bar{b}$ changed between orders, so there was **no new elimination**: the same A^{-1}, one multiplication per order.

Determinants and Inverses in Python

NumPy's `linalg` module

```
import numpy as np
A = np.array([[3, 1],
              [1, 2]])
np.linalg.det(A)
# 5.000000000000001
np.linalg.inv(A)
# [[ 0.4 -0.2]
#  [-0.2  0.6]]
b = np.array([5, 5])
np.linalg.solve(A, b)
# [1. 2.]

M = np.array([[2, 1],
              [2, 1]])
np.linalg.inv(M)
# LinAlgError: Singular matrix
```

Think with `inv`, compute with `solve`

- `solve` runs elimination directly on $A\bar{x} = \bar{b}$.
- It is faster, and loses less to rounding, than building `inv(A)` and multiplying.

Watch out

The trailing `...001` is floating point (Unit 1-5). Test whether `det` is **close** to 0, never equal.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §3 and §4

Exercise: Solve with the Inverse

Try it

1. Using the inverse $\begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$ from the last exercise, solve $2x + 5y = 1$, $x + 3y = 2$. Then solve the same left side with right-hand side $\begin{bmatrix} 4 \\ 2 \end{bmatrix}$.
2. The pharmacy is asked for 400 mL at 12 mg/mL. What does the formula $x = \frac{V(c-2)}{8}$, $y = \frac{V(10-c)}{8}$ give, and what does that mean?

Exercise: Solution

Answers

1. $\bar{x} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3-10 \\ -1+4 \end{bmatrix} = \begin{bmatrix} -7 \\ 3 \end{bmatrix}$. Check: $2(-7)+5(3) = 1$ and $-7+3(3) = 2$. With $\begin{bmatrix} 4 \\ 2 \end{bmatrix}$: $\begin{bmatrix} 12-10 \\ -4+4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. Check: $4+0 = 4$, $2+0 = 2$. One inverse, two systems.
2. $x = \frac{400(10)}{8} = 500$ mL and $y = \frac{400(-2)}{8} = -100$ mL. A **negative** volume: the order is impossible, because 12 mg/mL is stronger than the strongest stock. The algebra is fine; the request is not.

A First Look at Eigenvectors

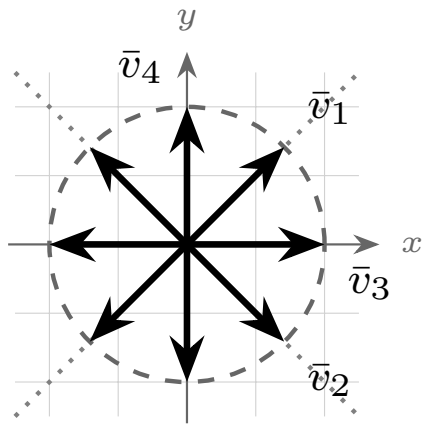
Directions That Do Not Rotate

Most matrices **scale and rotate** most vectors. But for many matrices, there are some directions that they **only scale**.

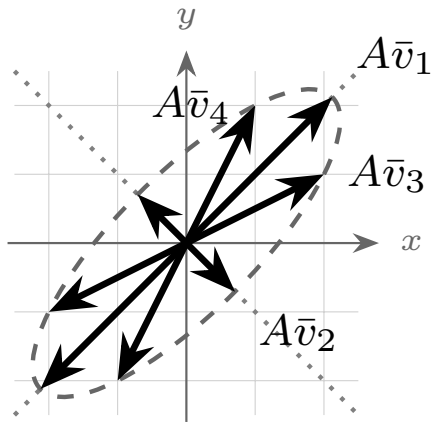
$$A = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

	$\bar{v}$	$A\bar{v}$	**direction**
$\bar{v}_1$	$\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$	$\frac{1.5}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$	**same**, $\times 1.5$
$\bar{v}_2$	$\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$	$\frac{0.5}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$	**same**, $\times 0.5$
$\bar{v}_3$	$\begin{bmatrix}1\\0\end{bmatrix}$	$\begin{bmatrix}1\\0.5\end{bmatrix}$	rotated
$\bar{v}_4$	$\begin{bmatrix}0\\1\end{bmatrix}$	$\begin{bmatrix}0.5\\1\end{bmatrix}$	rotated

Only $A\bar{v}_1$ and $A\bar{v}_2$ stay on the dotted lines; their negatives do too.



Inputs $\pm\bar{v}$, on the unit circle



Outputs $\pm A\bar{v}$, on the ellipse

Notebook `U2-6_Eigen-1_DirectionsThatDontTurn`, §1, most directions turn

Eigenvectors and Eigenvalues

Definition: eigenvector and eigenvalue

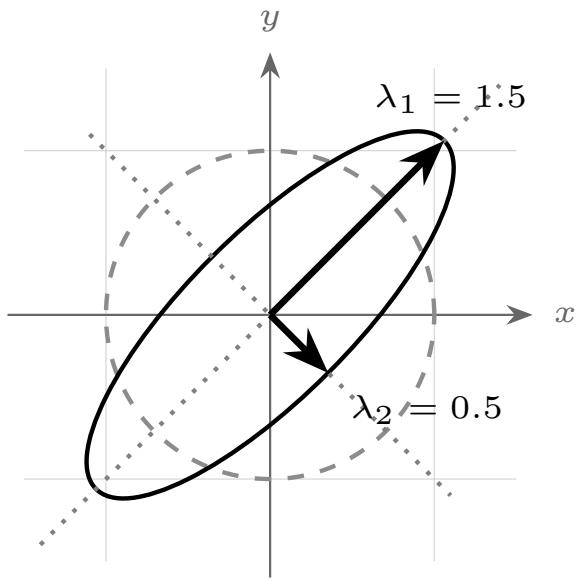
A **nonzero** vector $\bar{v}$ with

$$A\bar{v} = \lambda\bar{v}$$

is an **eigenvector** of A: it is only stretched, never turned. The factor λ is its **eigenvalue**. (*Eigen* is German for "own": A's own directions.)

From the previous slide

- $\bar{v}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$: $A\bar{v}_1 = 1.5\,\bar{v}_1$, so $\lambda_1 = 1.5$.
- $\bar{v}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$: $A\bar{v}_2 = 0.5\,\bar{v}_2$, so $\lambda_2 = 0.5$.
- $\bar{v}_3$, $\bar{v}_4$: rotated, so **not** eigenvectors.



Dashed: the unit circle. Solid: its image under A. The eigenvectors lie along the ellipse's axes.

Notebook `U2-6_Eigen-1_DirectionsThatDontTurn`, §3, the unit-circle animation

Finding the Eigenvalues: $\det(A - \lambda I) = 0$

The determinant does the work

$$
\begin{aligned}
A\bar{v} &= \lambda\bar{v} && \text{the definition} \\
A\bar{v} - \lambda I\bar{v} &= \bar{0} && \text{subtract } \lambda\bar{v}\text{, written as } \lambda I\bar{v} \text{ (since } I\bar{v} = \bar{v}\text{)} \\
(A - \lambda I)\,\bar{v} &= \bar{0} && \text{factor out } \bar{v}
\end{aligned}
$$

- **Question:** what would need to be true of $A - \lambda I$ for multiplying it by a **nonzero** $\bar{v}$ to give $\bar{0}$?
- **Answer:** it would need to be **singular**. (An inverse would undo it and force $\bar{v} = \bar{0}$.) So we can use the determinant: $\det(A - \lambda I) = 0$, the **characteristic equation**.

Notebook `U2-6_Eigen-2_CharacteristicEquation`, §1

Worked: the Eigenvalues of a 2×2

Solve $\det(A - \lambda I) = 0$ for $A = \left[\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right]$

$$A - \lambda I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

λI has λ only on the diagonal, so subtract λ there and nowhere else.

$$\begin{aligned} \det(A - \lambda I) &= (4-\lambda)(3-\lambda) - 1 \cdot 2 && ad - bc \\ &= 12 - 7\lambda + \lambda^2 - 2 && \text{expand} \\ &= \lambda^2 - 7\lambda + 10 && \text{collect terms} \\ &= (\lambda - 5)(\lambda - 2) = 0 && \text{factor} \end{aligned}$$

So the eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = 2$.

Why We Care About Eigenvectors

In data science

- **PCA.** Eigenvectors of the covariance matrix are the directions of **greatest spread**. Keep the top few to compress thousands of features into a handful.
- **PageRank.** A page's importance is its entry in an eigenvector of the link matrix.
- **Network communities.** Eigenvectors split a network into groups.

In medicine and science

- **Epidemics.** R_0 is the largest eigenvalue of the next-generation matrix; above 1, an outbreak grows.
- **Diffusion MRI.** The top eigenvector at each point gives the nerve fiber direction.
- **Populations.** The largest eigenvalue is the long-run growth rate.
- **Engineering.** A bridge's vibration modes are eigenvectors.

Common thread: apply a matrix again and again, and the **largest**-eigenvalue direction takes over.

How Many Eigenvalues?

The degree of the polynomial

- For an $n \times n$ matrix, $\det(A - \lambda I)$ is a polynomial of **degree** n: a quadratic for 2×2, a cubic for 3×3.
- So A has **at most** n eigenvalues: exactly n if you count repeats and complex ones.

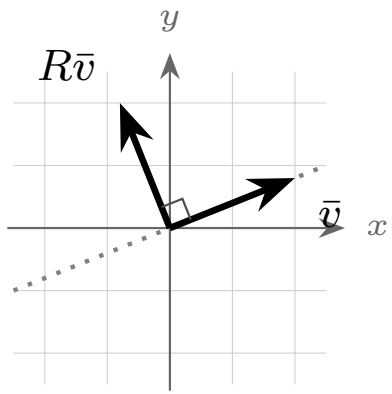
The rank of A

- **Rank**: the number of linearly independent columns (Unit 2-2).
- $\lambda = 0$ is an eigenvalue $\iff$ $\det A = 0 \iff$ rank $< n$.
- **Symmetric** A: rank = the number of **nonzero** eigenvalues.

A	$\det(A - \lambda I)$	**eigenvalues**	**rank**
$\left[\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right]$	$\lambda^2 - 7\lambda + 10$	$5,\ 2$	2
$\left[\begin{smallmatrix} 2 & 1 \\ 2 & 1 \end{smallmatrix}\right]$	$\lambda^2 - 3\lambda$	$3,\ 0$	1
$\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$	$\lambda^2 - \lambda$	$1,\ 0$	1
$\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$	$\lambda^2 + 1$	none real	2

Watch out: the last rule needs symmetry. $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ has eigenvalues $0, 0$ but rank 1.

Rotations: No Direction Survives



Every $R\bar{v}$ leaves the line of $\bar{v}$.

Intuition first: $R = R(90°) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

- R rotates **every** vector by $90°$.
- An eigenvector stays on its own line.
- None does: **no real eigenvectors**.

The algebra agrees, and says more

$$\det \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 + 1 = 0$$

$$\lambda^2 = -1$$

$$\lambda = \pm i$$

- No real number squares to -1, so we need $i = \sqrt{-1}$ from $\mathbb{C}$ (Unit 1-2).
- **Complex** eigenvalues: the algebra's way of saying "rotation."

Similar: any angle θ gives $\lambda = \cos\theta \pm i \sin\theta$, complex unless $\theta = 0°$ or $180°$. At $180°$, $R\bar{v} = -\bar{v}$: **every** vector is an eigenvector, $\lambda = -1$.

What the Eigenvalues Tell You

Their product is the determinant: three matrices from earlier slides

A	**eigenvalues**	**product**	$\det A = ad - bc$
$\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$	$1.5,\ 0.5$	$1.5 \cdot 0.5 = 0.75$	$1 - 0.25 = 0.75$
$\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$	$5,\ 2$	$5 \cdot 2 = 10$	$12 - 2 = 10$
$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	$i,\ -i$	$i \cdot (-i) = -i^2 = 1$	$0 - (-1) = 1$

The area factor is the product of the eigenvalues, even complex ones.

$\lambda = 0$ means singular: some direction is crushed to $\bar{0}$

- An eigenvalue of 0 makes the product 0, so $\det A = 0$ and A has no inverse.
- The projection $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$ has $\lambda = 1$ and $\lambda = 0$, and $\det = 1 \cdot 0 - 0 \cdot 0 = 0$.

Review: Determinants and Inverses

What we covered: the determinant

1. $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$ is the **signed area** of the parallelogram spanned by the columns: the factor by which A scales **every** area. In 3D it scales volume.
2. The **sign** records orientation: negative means flipped, as by a reflection.
3. The dot product measures the parallel part, $\|\bar{u}\|\|\bar{v}\| \cos\theta$; the determinant the perpendicular part, $\|\bar{u}\|\|\bar{v}\| \sin\theta$.
4. $\det = 0$ means three things at once: the plane **collapses** to a line, the columns are **dependent**, and A is **singular**. For a system, $\det \neq 0$ means exactly one solution for every right-hand side.
5. $\det(AB) = \det A \cdot \det B$: area factors multiply.

Review: Determinants and Inverses

What we covered: the inverse, and a first look at eigenvectors

6. A^{-1} **undoes** A: $A^{-1}A = AA^{-1} = I$. It exists exactly when $\det A \neq 0$, and $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Near $\det = 0$ the inverse blows up.
7. $(AB)^{-1} = B^{-1}A^{-1}$: undo the last move first.
8. $A\bar{x} = \bar{b}$ solves as $\bar{x} = A^{-1}\bar{b}$: invert once, reuse for every right-hand side. In code, prefer `np.linalg.solve`.
9. An **eigenvector** is only stretched, $A\bar{v} = \lambda\bar{v}$. Find λ from $\det(A - \lambda I) = 0$. The eigenvalues multiply to $\det A$. A rotation has complex eigenvalues: no real direction survives it.

Notebooks: `U2-5_Determinants-1`, `-2`; `U2-6_Eigen-1`, `-2`. **Previous:** Unit 2-4, Transformations. **Next:** Unit 3, **Calculus**.